Lesson 3: Strong walls for Teddy

Foundation STEAM | Houses unit | 45 minutes | CRT-ready

Overview	
Focus	The way we join materials together can make a house strong. Overlapping bricks makes a much stronger wall than stacking them in towers.
Curriculum	**VC2TDE2D03** use materials, components, tools and techniques to safely make designed solutions (joining bricks to build a strong wall). **VC2TDE2D04** describe and select design ideas and solutions based on personal preferences (testing the house and fixing weak parts).
Resources	**Lego tubs and base plates** (on the table next to the STEAM desk), **Teddy's letter** (page 5), STEAM journals and pencils. That is all.
Links	Students have looked at homes from the outside (Lesson 1) and the inside (Lesson 2). Now they meet Teddy, who needs a new house. Teddy's house is the thread through Lessons 4 to 7.

LISC

Learning Intention:

To learn that the way we join bricks together can make a house strong.

Success Criteria:

I can:

- overlap my bricks to build a strong wall.
- build a house with walls and a door with my partner.
- test my house and fix any parts that break.

Everything is ready for you

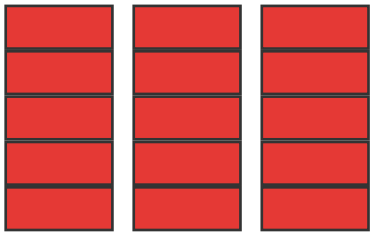
No preparation is needed. The **Lego tubs and base plates** are on the table next to the STEAM desk. Teddy's letter is with this plan. STEAM journals are in the class tub. Bring the class straight to the mat and grab a handful of Lego bricks for the demonstration.

Lesson sequence

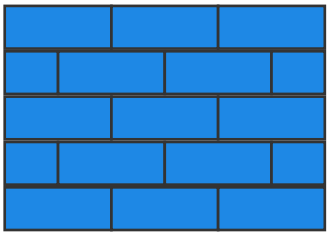
Time	What to do
0 to 5 min **Hook (CIC)**	**A letter from Teddy.** Read Teddy's letter aloud with a sad voice. Ask: *"What happened to Teddy's house? Why did it fall down?"* Let a few students answer. Then: *"Teddy needs our help. Today we are going to be builders and find out how to make walls that do not fall down."*
5 to 12 min **Wall test**	On the mat, build two short walls in front of the class (see the picture below). **Wall A:** stack bricks straight up in 3 separate towers, side by side. **Wall B:** overlap the bricks, like a real brick wall. Ask: *"Which wall do you think is stronger?"* Students point. Then gently push the top of each wall. Wall A breaks apart; Wall B holds together. Ask: *"Why did Wall B stay together?"* Finish with the big idea, said together: **"Overlap the bricks to make a strong wall."** If there is a brick wall visible from the room, point it out.
12 to 14 min **Builder rules**	Explain the job: *"With your partner, build a house for Teddy on a base plate. It needs 4 walls with overlapping bricks and a door. If you finish, add a roof."* Rules: share the bricks, build on the base plate only, and no breaking other people's houses. Put students in pairs and send each pair to a table. Helpers put out one base plate per pair and one Lego tub per table.
14 to 28 min **Build**	Students build. Walk around and ask: *"Show me where you overlapped your bricks. How tall is your wall? Count the bricks."* If you see towers instead of overlapping, point to Wall B and ask: *"Does your wall look like this one?"*
28 to 33 min **Wobble test**	Stop the class. Tell them a big wind is coming. On your count, each pair holds their base plate **low over the table** and gently wobbles it while everyone counts to 5. Then: *"Did anything fall off? Which part broke?"* Give pairs 2 minutes to **fix the weak part**, then do the wobble test again.
33 to 38 min **Journal**	Each student draws their house in their STEAM journal and draws a circle around the strongest part. Walk around and write a short label for them in their words, for example *"Strong because we overlapped."*
38 to 41 min **Share and reflect**	Choose 2 or 3 pairs to show their house and say how they made it strong. Read the Success Criteria aloud and students show a thumb (up, sideways or down) for each one. Teaser: *"Next week we will find out how to keep Teddy dry in the rain."*
41 to 45 min **Pack up**	Houses are pulled apart and all bricks go back in the tubs. Base plates are stacked on the Lego table. Journals go back in the class tub.

The wall test: what to build

Use the same size bricks for both walls, about 3 bricks wide and 5 bricks tall. In Wall A the bricks sit straight on top of each other, so each column is a separate tower. In Wall B each brick sits across the join of the two bricks below it.



Wall A: towers (weak)



Wall B: overlapping (strong)

Questions to ask

- Where did you overlap your bricks?
- How many bricks tall is your wall?
- What happened in the wobble test? Which part broke?
- How did you fix it? Is it stronger now?
- Real builders overlap bricks too. Can you think of a brick wall you have seen?

What to look for

Success criterion	Evidence of learning
Overlaps bricks	Walls show bricks sitting across joins, not separate towers.
Builds a house with a partner	House on the base plate has 4 walls and a door; partners share bricks and take turns.
Tests and fixes	After the wobble test, can point to the part that broke and show how they made it stronger.

Support and extension

Level	Idea
Support	Build the first row of the wall together with the pair, then let them continue. Give the sentence starter: *"My house is strong because ___."*
Extension	Give these challenges out loud to pairs who finish early: **add a window**; **add a roof** that covers the whole house; **make it tall enough** for a Lego person to stand inside; **build a second floor**. After each one, do another wobble test.

Notes for the CRT

Keep the wobble gentle. Base plates stay low over the table so nothing drops on the floor. **Leave 4 minutes** to pack up: houses are pulled apart and every brick goes back in the tubs, so nothing needs to be stored. Please leave the **STEAM journals** in the class tub on the STEAM desk. A short note about how the class went is always appreciated.

Dear Foundation,

I need your help! I built my house, but when the big wind came, my walls wobbled and my house fell down.

Now I have nowhere to live. Can you help me build a house with strong walls that will not fall down?

Love from,
Teddy